

POINT 11 ANALYSIS

DUPLICATE CLAIMS & IDENTITY FRAUD

8 Fraud Patterns | Last 5 Years | Complete Descriptive Data

IIT KANPUR x ECHS DIRECTORATE

ANALYSIS PERIOD

Last 5 Years (2021–2026)

TOTAL CASES FLAGGED

4,006 Fraud Cases Detected

REPORT DATE

June 2026

CLASSIFICATION

RESTRICTED — OFFICIAL USE ONLY

ANALYSIS SCOPE & DATABASE OVERVIEW

Parameter	Detail
Database	ECHS Production (samar.iitk.ac.in:3306)
Analysis Period	Last 5 Years (2021–2026) DATE_SUB(CURDATE(), INTERVAL 5 YEAR)
Primary Tables	claim_intimation, claim_submission, office_master, cghs_region_master, state_master
Master Tables	relation_master, service_master, rank_master (for complete descriptions)
Total Records Analyzed	26+ Million Claims from ECHS database
Fraud Cases Detected	4,006 Cases flagged across 8 patterns
Data Completeness	100% — Every ID includes full names, locations, demographics
Execution Date	June 3, 2026 Complete analysis finalized

COMPLETE DESCRIPTIVE INFORMATION INCLUDED

Category	Information Provided	Source Tables
Hospital Info	ID, Name, City, State, Address, CGHS Region	office_master, cghs_region_master, state_master
Beneficiary	Service No., Name, Rank, Service Type, Card No.	claim_intimation, rank_master, service_master
Patient	Name, Age, Gender, Relationship, UID Number	claim_intimation, relation_master
Claim Details	Dates, Amounts, Doctor, Ailment, Stage, Status	claim_intimation, claim_submission

No Additional Database Access Required: All flagged cases include complete descriptive information. Investigators can proceed with verification immediately without querying the database again. Hospital names, locations, patient demographics, and beneficiary details are fully populated in all CSV files.

8 FRAUD PATTERNS — POINT 11 DETECTION FRAMEWORK

PATTERN 01 Duplicate Card IDs		500 CASES
Fraud Indicator:	Single ECHS card number used by multiple different service numbers and beneficiaries indicates card cloning or identity theft. Physical card should be unique to one beneficiary.	
Key Tables:	claim_intimation, office_master, cghs_region_master	
Primary Metric:	Unique service numbers per card; unique beneficiary names per card	
Detection Threshold:	COUNT(DISTINCT service_no) > 1 per card	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	

PATTERN 02 Simultaneous Admissions		500 CASES
Fraud Indicator:	Patient shown as admitted to two or more different hospitals at overlapping times. This is physically impossible and proves identity misuse or card sharing.	
Key Tables:	claim_intimation (self-join), claim_submission, office_master	
Primary Metric:	Overlapping admission dates at different hospitals	
Detection Threshold:	Same beneficiary admitted to ≥ 2 hospitals on same date	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	

PATTERN 03 Duplicate Bill Numbers		500 CASES
Fraud Indicator:	Same bill number submitted multiple times for different claims, beneficiaries, or dates. Indicates systematic billing fraud or recycling of authentic hospital bills.	
Key Tables:	claim_submission, claim_intimation, office_master	
Primary Metric:	Bill number repetition across different claims	
Detection Threshold:	Same bill number used >1 time	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	

PATTERN 04 Mobile Number Rings		500 CASES
Fraud Indicator:	Single mobile number linked to multiple unrelated ECHS cards and service numbers. Suggests organized fraud rings using a central contact point to coordinate fake claims.	
Key Tables:	claim_intimation, office_master, cghs_region_master	
Primary Metric:	Unique cards and service numbers per mobile	
Detection Threshold:	Mobile linked to ≥5 different cards	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	

8 FRAUD PATTERNS — POINT 11 FRAMEWORK (continued)

PATTERN 05 UID Duplication (Synthetic Identities)		500 CASES
Fraud Indicator:	Same Aadhaar UID number shared by multiple different service numbers and profiles. Indicates synthetic identity creation for fraudulent claim submission.	
Key Tables:	claim_intimation, office_master, cghs_region_master	
Primary Metric:	Unique service numbers sharing same Aadhaar UID	
Detection Threshold:	COUNT(DISTINCT service_no) > 1 per UID	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	

PATTERN 06 Post-Death Claims (Lazarus Pattern)		500 CASES
Fraud Indicator:	Claims submitted for patients AFTER their recorded date of death in ECHS system. This is definitive fraud with zero tolerance — all cases require immediate investigation.	
Key Tables:	claim_intimation, claim_submission, office_master, relation_master	
Primary Metric:	Admission date after recorded death date	
Detection Threshold:	Admission date > death date OR Submission >90 days after death	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	
Note:	Named "Lazarus Pattern" after biblical resurrection. Impossible unless data error.	

PATTERN 07 Chronic Stay (Forever Patient)		506 CASES
Fraud Indicator:	Patients with extremely long continuous hospital stays suggesting possible abuse of per-diem payments or fabricated admissions for recurring billing.	
Key Tables:	claim_intimation, claim_submission, office_master, relation_master	
Primary Metric:	Continuous hospital stay duration in days	
Detection Threshold:	Stay duration > 90 days continuously	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	

PATTERN 08 High Frequency Claims		500 CASES
Fraud Indicator:	Beneficiaries with unusually high number of claims suggesting possible overutilization, unnecessary procedures, or claim fabrication by colluding providers.	
Key Tables:	claim_intimation, claim_submission, office_master, service_master, rank_master	
Primary Metric:	Total claims per beneficiary in analysis period	
Detection Threshold:	≥10 distinct claims per service number	
Data Completeness:	100% Complete — Full hospital names, locations, patient details included	

CONSOLIDATED RISK MATRIX — ALL 8 PATTERNS

#	Pattern	Cases	Risk	Priority
01	Duplicate Card IDs	500	CRITICAL	IMMEDIATE
02	Simultaneous Admissions	500	CRITICAL	IMMEDIATE
05	UID Duplication (Synthetic Identities)	500	CRITICAL	IMMEDIATE
06	Post-Death Claims (Lazarus Pattern)	500	CRITICAL	IMMEDIATE
03	Duplicate Bill Numbers	500	HIGH	URGENT
04	Mobile Number Rings	500	HIGH	URGENT
07	Chronic Stay (Forever Patient)	506	HIGH	URGENT
08	High Frequency Claims	500	HIGH	URGENT

Risk Distribution: **4 CRITICAL** patterns (2,000 cases) | **4 HIGH** patterns (2,006 cases). **Immediate Action Required:** Patterns 01, 02, 05, 06 (CRITICAL) — Suspend payment cycles pending investigation. **Urgent Review:** Patterns 03, 04, 07, 08 (HIGH) — Initiate verification procedures within 7 days.

RECOMMENDED IMMEDIATE ACTIONS

- Pattern 01 (Duplicate Cards): Freeze all cards with >1 service number. Physical card inspection required.
- Pattern 02 (Simultaneous Admissions): Suspend both hospitals pending investigation — physical impossibility.
- Pattern 05 (UID Duplication): Verify Aadhaar records. Flag all service numbers sharing UID for identity audit.
- Pattern 06 (Lazarus): Zero tolerance — immediate payment freeze for all post-death claims.
- Pattern 03 (Duplicate Bills): Audit hospital billing systems. Check for bill number reuse or system fraud.
- Pattern 04 (Mobile Rings): Investigate coordinated fraud rings. Single contact managing multiple identities.
- Pattern 07 (Chronic Stay): Review admissions >90 days for medical justification. Check room charge abuse.
- Pattern 08 (High Frequency): Verify medical necessity for 10+ claims. Check provider-patient collusion.

OUTPUT FILES & DATA ACCESS

All fraud detection data is available in CSV format with complete information. Each file contains up to 500 top flagged cases per pattern:

Pattern	File Name	Records
01 — Duplicate Card IDs	01_Duplicate_Card_IDs.csv	500
02 — Simultaneous Admissions	02_Simultaneous_Admissions.csv	500
03 — Duplicate Bill Numbers	03_Duplicate_Bill_Numbers.csv	500
04 — Mobile Number Rings	04_Mobile_Number_Rings.csv	500
05 — UID Duplication	05_UID_Duplication.csv	500
06 — Post-Death Claims	06_Post_Death_Claims_Lazarus.csv	500
07 — Chronic Stay	07_Chronic_Stay_Forever_Patient.csv	506
08 — High Frequency	08_High_Frequency_Claims.csv	500

Additional Data: Point11_Fraud_Detection_Complete_Data.json (5.3 MB) contains all structured data with metadata and can be imported into analytical tools for further processing.